The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Decisions

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A controller that acts on delayed information, through a sensing lag, a telemetry delay or a stale cache, can beat the best static policy only insofar as the delayed observation still predicts the current state. A companion systems submission [1] proves this exactly for symmetric two-state Markov capacity. This paper supplies the general theory. It isolates the exact quantity, the Bayes envelope, which is the expected excess value of posterior over prior beliefs, and restates in value-of-information form an envelope known from the minimum-excess-risk literature that bounds it by the square root of the mutual information between the delayed observation and the current state. Three consequences follow. An observation carrying a fixed number of bits caps the value of feedback whatever it encodes. The envelope is tight to leading order for the symmetric binary chain. No universal formula can map mutual information to decision value, because two observations with identical information can carry all or none of the decision-relevant distinction. For binary state, the local decision geometry at the prior sets how value grows with correlation: as a sum of hinges with dead zones for finite action sets, and quadratically when the value function is smooth. A ten-series battery across acoustics, bioacoustics, space weather, seismology and the BGP control plane of the Internet, with synthetic controls, tests these predictions. Two series, a teleseismic earthquake record and a BGP update stream, carry material history value, and calendar blocks show that the value is concentrated in identifiable regimes. A chain-rule decomposition separates what history reveals about the regime from what it predicts within one, and the envelope inverts into dependence-aware lower bounds on conditional predictive information. All history values are measured for a protocol-fixed six-lag lookup policy and are lower bounds on what a richer policy could extract.

Additional Key Words and Phrases: value of information, age of information, Bayes envelope, minimum excess risk, mutual information

1 Introduction

Everywhere a controller acts on stale state (a scheduler admitting jobs against capacity sensed seconds ago, a trader on delayed quotes, a relay on an outdated channel estimate), the same question recurs: *what is the value of information of age D?* The age-of-information literature [7] measures freshness; freshness, however, has no decision semantics: a maximally fresh sample of an i.i.d. process is worthless, and a stale sample of a slowly-mixing one is nearly as good as the present. A companion systems paper [1] made this precise in one setting, proving that for symmetric two-state Markov capacity the advantage of age-D feedback over the best static policy is *exactly* $\Delta(D) = \frac{1}{2}r(D)$ (with $r(D)$ the autocorrelation at the sensing lag), and measuring the law on a live GPU cluster. This paper supplies the theory that result belongs to. The thesis is one line:

Age matters only through predictive information, and predictive information matters only through the Bayes envelope.

Concretely: (**C1**) we isolate the exact object, the Bayes envelope $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, classical in decision theory [2–4], and restate in that frame the *predictive-information envelope*

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$\Delta_Y \le \Omega\sqrt{I(X;Y)/2}$ (Theorem 3.1; known as a minimum-excess-risk inequality, attribution in §3), with the *bit-budget corollary*: an observation carrying at most b bits ($H(Y) \le b \ln 2$) caps feedback value at $\Omega\sqrt{(b \ln 2)/2}$, so compression can preserve but never create feedback value. For integer b, the cap improves on the trivial range bound only when $b \le 2$, and a controller that accumulates messages across decisions holds a richer observation to which the corollary applies with its total entropy. (**C2**) We show the envelope is *sharp where it matters*: on the canonical binary chain it matches the exact law to $O(r^3)$ (Proposition 3.6), and the battery of §7 *illustrates* this tightness on real binary pair distributions (a consistency check: $\hat{I}$ and $\widehat{V}_1$ derive from the same lag table). (**C3**) We prove no universal equality $\Delta = f(I)$ exists for bounded objectives (Proposition 4.1): information is a scalar, value is a geometry. This stands in contrast to the classical exception. For *logarithmic* (unbounded) utility, Kelly and Barron–Cover [5, 6] show the value of side information *equals* I. For general bounded objectives, mutual information provides a universal envelope but does not determine value; logarithmic utility is a distinguished objective for which equality is possible. (**C4**) We give an exactness theory for binary state whose organizing fact is that *the local geometry of the value function at the prior sets the correlation exponent.* For finite action sets (polyhedral value functions) the value of a delayed sample is exactly a sum of *hinges* in the lag-D covariance, with sign-dependent dead-zone thresholds (linear response beyond, identically zero within); when the value function ψ is twice differentiable at the prior with $\psi'' > 0$, the weak-correlation response is quadratic to leading order (exactly $C(D)^2/\pi(1-\pi)$ for squared error); and a prior sitting on a kink (the companion's symmetric case) is the polyhedral configuration yielding threshold-free linear response (§6). The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [18, 19]: when the prior sits off a kink of a polyhedral objective, weakly-correlated feedback is *worthless*, not merely weak, and the sign of the correlation matters. (**C5**) We order the three envelopes (mixing, information, bit-budget) below the master total-variation envelope and prove mixing and information are incomparable. For every lag and every K, a *preview process* keeps $\beta(D) \ge \frac{1}{2}$ while the information envelope, and the true value, vanish with K (§5). (**C6**) Across ten series from four physical domains, one computer system and controls, the battery establishes equivalence for the protocol-fixed policy class on three series, leaves one noise-limited and one inconclusive, records two smaller detections near the practical threshold, and finds material history gains on two, a teleseismic record and a BGP update stream; the i.i.d. series serves as the selection control. Calendar blocks show both gains concentrated in identifiable regimes, which a chain-rule decomposition (Lemma 3.5) separates into regime identification and within-regime prediction. Read backwards, the incremental theorem converts each gain into a model-calibrated, dependence-aware lower bound on *conditional* history information, always one-sided: $I(X_t; \text{history} \mid X_{t-D}) \ge 2[(\widehat{\Delta}_W - \widehat{\Delta}_Z - \varepsilon)^+]^2$ (§7). Every history value in the battery is measured for a protocol-fixed six-lag lookup policy, so it is a lower bound on what a richer policy (longer lags, a recurrent model) could extract, and every negative finding is a statement about that policy class.

Table 1 separates what this paper restates from what it adds. The finite and algebraic results are machine-checked in Lean 4 against Mathlib (52 theorems), and the analytic ones symbolically; both are part of the released artifact.

The scope is stated precisely: this is a unified theory for *exogenous one-step bounded-reward decisions* under delay. The action does not move the state, and the reward is single-stage. Within that scope the theory is closed on itself: one decision-theoretic object explaining an exact systems law, envelopes ordered by what they require (a model, samples, or a bit rate), sharpness results tying envelope to law, impossibility results delimiting both, and a falsifiable empirical program run on real processes from four physical domains and one computer system.

Component	Status
Bayes envelope (1)	classical [2–4]
Information envelopes (Thm. 3.1, Thm. 3.4)	known minimum-excess-risk inequalities, restated
Garbling monotonicity (Lemma 3.3)	classical (Blackwell)
Bit-budget corollary (Cor. 3.2)	immediate from $I(X;Y) \le H(Y)$
No universal $\Delta = f(I)$ (Prop. 4.1)	elementary construction
Local marginal-value trichotomy	known [3]
Exact lag-covariance hinge law, sign-asymmetric thresholds (Thm. 6.1)	new, to our knowledge
Explicit contact-order coefficients (Thm. 6.3)	new, to our knowledge
Mixing/information separation construction (Prop. 5.2)	new, to our knowledge
Regime decomposition (Lemma 3.5)	elementary (chain rule)
Conditional-information certificate from policy gain	new operational use
Multi-domain battery with a BGP series and regime analysis (§7)	new

Table 1. What is known and what is new. The known results are restated because the exact laws and the battery are built on them.

2 Setting and the exact object

Let X be the current exogenous state taking values in a finite set $\mathcal{X}$, with marginal $\bar{\mu}$. A controller observes a delayed quantity Y (a delayed sample X_{t-D}, a delayed history, or any $\sigma(\mathcal{F}_{\le t-D})$-measurable statistic), chooses an action w, and receives a bounded one-step reward $R(w,x)$ with oscillation $\Omega = \sup_{w,x} R - \inf_{w,x} R$. Define the value of a belief $\mu \in \Delta(\mathcal{X})$:

$$\psi(\mu) \;=\; \sup_{w} \sum_{x} R(w,x)\,\mu(x),$$

We assume throughout that the supremum is attained by a measurable policy, which is automatic for finite action sets and a standing assumption for continuous ones. The function ψ is a pointwise supremum of linear functionals and hence convex. Let $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$ denote the posterior; by the tower property, $\mathbb{E}[\mu_Y] = \bar{\mu}$. The optimal Y-measurable controller attains $V_Y^\star = \mathbb{E}[\psi(\mu_Y)]$; the best static controller attains $V_{\text{stat}} = \psi(\bar{\mu})$. The exact object of this paper is the **Bayes envelope**

$$\Delta_Y \;=\; V_Y^\star - V_{\text{stat}} \;=\; \mathbb{E}\big[\psi(\mu_Y)\big] - \psi(\bar{\mu}) \;\ge\; 0, \tag{1}$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

Lemma 2.1 (Lipschitz value; standard, used in [1, App. A]). $|\psi(\mu) - \psi(\mu')| \le \Omega\,\|\mu - \mu'\|_{\text{TV}}$, *hence* $\Delta_Y \le \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}}$.

The proof is two lines and is given in Appendix A. The total-variation route specializes to the β-mixing converse of [1] when Y is the full delayed history: $\Delta(D) \le \Omega\,\beta(D)$.

3 The predictive-information envelope

Theorem 3.1 (Predictive-information bound; restated from [12, 13]). *For any bounded objective with oscillation Ω and any delayed observable Y,*

$$0 \;\le\; \Delta_Y \;\le\; \Omega\sqrt{\tfrac{1}{2}\,I(X;Y)} \qquad (I \text{ in nats; in bits, } \Omega\sqrt{\tfrac{\ln 2}{2}\,I_{\text{bits}}}). \tag{2}$$

Proof. By Lemma 2.1, $\Delta_Y \le \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$. Pinsker's inequality gives, for every realization y, $\|\mu_y - \bar{\mu}\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\mu_y \| \bar{\mu})}$. Taking the expectation over Y and applying Jensen to the concave square root,

$$\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}} \;\le\; \mathbb{E}\sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(\mu_Y \| \bar{\mu})} \;\le\; \sqrt{\tfrac{1}{2}\,\mathbb{E}\, D_{\mathrm{KL}}(\mu_Y \| \bar{\mu})} \;=\; \sqrt{\tfrac{1}{2}\, I(X;Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior. □

Corollary 3.2 (Bit-limited telemetry). *Suppose the delivered observation satisfies $H(Y) \le b \ln 2$ nats (in particular, whenever Y takes at most 2^b values, or is the output of a prefix code of expected length $\le b$ bits). Then*

$$\Delta_Y \;\le\; \Omega\, \min\Big\{1,\ \sqrt{(b \ln 2)/2}\Big\},$$

the clip because the trivial range bound $\Delta_Y \le \Omega$ beats the square root already at $b \ge 3$ bits.

Lemma 3.3 (Garbling never helps: value monotonicity). *If $Y = g(Z)$ (deterministically or through any channel acting on Z alone), then $\Delta_Y \le \Delta_Z$, and also $\Delta_Y \le \Omega\sqrt{\frac{1}{2} I(X;Z)}$.*

Proof. $\mu_Y = \mathbb{E}[\mu_Z \mid Y]$ by the tower property, so convexity of ψ and Jensen give $\mathbb{E}\psi(\mu_Y) \le \mathbb{E}\psi(\mu_Z)$; subtract $\psi(\bar{\mu})$. The second claim is Theorem 3.1 plus data processing ($I(X;Y) \le I(X;Z)$). Note the two statements are different in kind: data processing alone bounds only the information *envelope*; the Jensen step bounds the *value itself*. This is the Blackwell comparison-of- experiments order [17] specialized to garbled delayed sensing. Together: compression can preserve feedback value; no encoding of the delayed signal can create it. □

The same tower-property mechanism yields the theorem that governs *richer* observations, the form §7 actually needs:

Theorem 3.4 (Incremental-information envelope). *Let Z be any delayed observable and $W = (Z, H)$ a refinement (e.g. $Z = X_{t-D}$ a single delayed sample, H the rest of the delayed history window). Then*

$$0 \;\le\; \Delta_W - \Delta_Z \;\le\; \Omega\sqrt{\tfrac{1}{2}\, I(X; H \mid Z)}.$$

Equivalently, read backwards: incremental decision value certifies conditional predictive information,

$$I(X; H \mid Z) \;\ge\; 2\Big(\frac{\Delta_W - \Delta_Z}{\Omega}\Big)^2.$$

Proof. $\mu_Z = \mathbb{E}[\mu_W \mid Z]$ by the tower property, so $\Delta_W - \Delta_Z = \mathbb{E}[\psi(\mu_W) - \psi(\mu_Z)] \ge 0$ by Jensen (Lemma 3.3), and pointwise $\psi(\mu_W) - \psi(\mu_Z) \le \Omega\|\mu_W - \mu_Z\|_{\mathrm{TV}}$ (Lemma 2.1). Pinsker and Jensen then give $\mathbb{E}\|\mu_W - \mu_Z\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\,\mathbb{E}\, D_{\mathrm{KL}}(\mu_W \| \mu_Z)} = \sqrt{\frac{1}{2}\, I(X; H \mid Z)}$, the last equality being the definition of conditional mutual information as the expected KL divergence of the refined posterior from the coarse one. □

Theorem 3.1 is the case $Z = \varnothing$. The pair separates two questions a delayed sensor can ask: *how much is the sample worth?* (Theorem 3.1 at $Y = Z$) and *how much more is the history worth?* (Theorem 3.4). The second is bounded by information the single sample does *not* carry, never by the single-sample information itself.

Attribution. Neither inequality is new. In the minimum-excess-risk literature the unconditional form appears with a weaker constant in [11] and in the present form in [12] (Cor. 1 and Rem. 3(ii); Rem. 3(iii) notes the Pinsker proof) and [13] (Thm. 1 and Cor. 5, with the oscillation constant); the conditional/incremental form is [15] (Lem. 1) and [14] (Cors. 2–3), and the Lipschitz–Pinsker–Jensen engine is [16] (Fact 9 and Prop. 4). What this paper claims is the value-of-delayed-information reading, the corollaries in that reading (bit budget, Blackwell garbling), the operational inversion the battery executes (§7), and the exactness theory that anchors the envelopes (§6).

Lemma 3.5 (Regime decomposition). *For any delayed observables Z, H and any random variable R (a regime label),*

$$I(X;H \mid Z) \;\le\; I(X;R \mid Z) + I(X;H \mid Z,R),$$

with equality when R is a function of (Z,H). Hence the incremental value of Theorem 3.4 satisfies $\Delta_W - \Delta_Z \le \Omega\sqrt{\tfrac{1}{2}[I(X;R \mid Z) + I(X;H \mid Z,R)]}$.

Proof. By the chain rule, $I(X;H,R \mid Z) = I(X;R \mid Z) + I(X;H \mid Z,R)$, and $I(X;H \mid Z) \le I(X;H,R \mid Z)$, with equality when R is determined by (Z,H). □

The lemma is elementary, and we state it because it is what the battery measures. A history gain pooled over a nonstationary record bounds the sum, so part of it can be regime identification: history that only reveals which regime the system is in. A history gain measured inside one regime bounds the within-regime term. The two readings coincide only for a stationary process, and §7 measures both.

Proposition 3.6 (Leading-order tightness in the canonical case). *For a stationary fair binary pair with lag-D correlation $r = r(D)$,*

$$I(X_t;X_{t-D}) = \tfrac{1}{2}\big[(1{+}r)\ln(1{+}r) + (1{-}r)\ln(1{-}r)\big] = \tfrac{r^2}{2} + \tfrac{r^4}{12} + O(r^6),$$

an even function of r, and the exact single-sample value of the $0/1$ prediction objective is $\Delta(D) = \tfrac{1}{2}|r(D)|$ (negative correlation is exploited by reversing the prediction). The right side of (2) equals $\Omega\big(\tfrac{|r|}{2} + O(|r|^3)\big)$: the envelope matches the exact law to leading order at $\Omega = 1$. (The companion's persistent-chain assumption (flip probability $\le \tfrac{1}{2}$) gives $r \ge 0$ and its statement $\tfrac{1}{2}r$; we carry the absolute value throughout.) The information envelope is not loose where it matters.

Proof. The joint distribution places mass $\frac{1+r}{4}$ on each agreeing pair and $\frac{1-r}{4}$ on each disagreeing pair; the stated I follows by direct computation, and the expansion from $(1 \pm r)\ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \cdots$. Then $\sqrt{\frac{1}{2}I} = \frac{|r|}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)}$. □

4 No universal equality $\Delta = f(I)$

Proposition 4.1 (Information is a scalar; value is a geometry). *There is a state space, a bounded objective, and two observations Y_1, Y_2 with $I(X;Y_1) = I(X;Y_2) > 0$ but $\Delta_{Y_1} = \frac{\Omega}{2} > 0 = \Delta_{Y_2}$. Hence no function f with $\Delta_Y = f(I(X;Y))$ exists for all experiments.*

Proof. Let $X = (B_1, B_2)$ be two independent fair bits and let the objective depend only on B_1: actions $w \in \{0,1\}$, $R(w,x) = \Omega\,\mathbf{1}[w = B_1]$ (so the oscillation is Ω). Static value: $\Omega/2$. Let $Y_1 = B_1$ and $Y_2 = B_2$. Both have $I(X;Y_i) = \ln 2$. Observing Y_1 yields $V^\star = \Omega$, so $\Delta_{Y_1} = \Omega/2$; observing Y_2 leaves the posterior over B_1 uniform, so $\Delta_{Y_2} = 0$. The two experiments share every information quantity in $I(X;Y)$ yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other. □

Remark 4.2. *This is the precise sense in which the companion paper's binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains* (1).

5 Envelope comparison

The three envelopes are not interchangeable, and their relationship is an ordered pair of relaxations of one master quantity.

Throughout, we use the companion's event-supremum convention

$$\beta(D) \;=\; \mathbb{E}\Big[\sup_{B\in\mathcal{F}_{\geq t}} \big|\mathbb{P}(B \mid \mathcal{F}_{\leq t-D}) - \mathbb{P}(B)\big|\Big],$$

which lower-bounds the absolute-regularity coefficient; conventions differ by factors of two across the literature, and every displayed constant below is tied to this one. Lower bounds on this β are what vacuousness of the mixing envelope requires.

Proposition 5.1 (Master envelope and universal ordering). *For any delayed observable Y measurable w.r.t. the delayed history $\mathcal{F}_{\leq t-D}$,*

$$\Delta_Y \;\leq\; \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}} \;\leq\; \Omega\,\min\Big\{\beta(D),\ \sqrt{\tfrac{1}{2}I(X;Y)}\Big\},$$

where $\beta(D)$ is the β-mixing coefficient of the state process at lag D. The expected posterior movement $\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$ is the master envelope; the mixing and information envelopes are its two relaxations.

Proof. The first inequality is Lemma 2.1. For the second: since Y is $\mathcal{F}_{\leq t-D}$-measurable and $\sigma(X) \subseteq \mathcal{F}_{\geq t}$, conditioning on the coarser Y can only shrink expected posterior movement, and $\mathbb{E}\|\mu_{\mathcal{F}_{\leq t-D}} - \bar{\mu}\|_{\mathrm{TV}} \leq \beta(D)$ because the β-coefficient takes its supremum over the larger future field [1, App. A]. The information relaxation is the Pinsker step in the proof of Theorem 3.1. □

Proposition 5.2 (The two relaxations are incomparable). *Neither relaxation dominates the other.*

(i) Mixing strictly tighter: *for the symmetric two-state chain with $Y = X_{t-D}$, $\beta(D) = \frac{1}{2}|r|$ while $\sqrt{\frac{1}{2}I} = \frac{|r|}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)} > \beta(D)$ for every $|r| \in (0,1)$.*

(ii) Information arbitrarily tighter: *for every fixed lag D and every $K \geq 2$ there is a stationary state process and bounded objective with*

$$\beta(D) \;\geq\; \tfrac{1}{2}, \qquad I(X_t; \mathcal{F}_{\leq t-D}) \;\leq\; h_2(1/K) + \tfrac{\ln 2}{K} \xrightarrow[K\to\infty]{} 0, \qquad \Delta \;=\; \tfrac{\Omega}{2K},$$

where h_2 is the binary entropy in nats: the mixing envelope stays $\geq \Omega/2$ while the information envelope, and the true value, vanish.

Proof. (i) is Proposition 3.6 together with $\beta(D) = \frac{1}{2}|r(D)|$ for the binary chain (Appendix A). (ii) *The preview process (one per (D,K)).* Fix D and K, set $m = D + K$. Let c_t be i.i.d. fair bits, let $\Theta \sim \mathrm{Unif}\{0,\dots,K-1\}$ be an independent phase (randomized to make the pair stationary), and let the observable state be $X_t = (c_t, p_t)$ with

$$p_t \;=\; \begin{cases} c_{t+m} & t \equiv \Theta \pmod K \\ \perp & \text{otherwise,} \end{cases}$$

a sensor that occasionally publishes a *forecast* m steps ahead. The objective depends only on the current payoff bit: $R(w,x) = \Omega\,\mathbf{1}[w = c]$. *Mixing (lower bound).* The window $[t-m,\ t-D-1]$ contains K consecutive integers, hence exactly one preview slot $s \equiv \Theta$; that preview is in the delayed past and reveals c_{s+m} with $s + m \in [t,\ t+K-1]$, a present-or-future payoff bit. The event

$B = \{c_{s+m} = 1\} \in \mathcal{F}_{\geq t}$ then has conditional probability in $\{0, 1\}$ against marginal $\frac{1}{2}$, so $\beta(D) \geq \frac{1}{2}$. (Previews shared between past and future create further dependence; we claim only the lower bound, which is what vacuousness of the mixing envelope needs.) *Information (upper bound).* X_t is determined by the triple (payoff bit c_t; slot indicator $\mathbf{1}[t \equiv \Theta]$; preview value when present, a *future* bit independent of the past). The delayed history reveals the slot indicator (it determines Θ) and reveals c_t itself exactly when $t - m \equiv \Theta$ (probability $1/K$); hence $I(X_t; \mathcal{F}_{\leq t-D}) \leq h_2(1/K) + \frac{\ln 2}{K}$. *Value.* Feedback wins $\Omega/2$ on the revealed fraction $1/K$ and nothing elsewhere: $\Delta = \Omega/(2K)$. The information envelope $\Omega\sqrt{(h_2(1/K) + \ln 2/K)/2}$ decays with K (rate $\widetilde{O}(K^{-1/2})$ against the true K^{-1}, loose but not blind); the mixing envelope does not decay at all. □

Remark 5.3 (Why keep the information form at all). *Proposition 5.1 shows the TV master envelope is always at least as tight, and (i) shows mixing can beat information. Three properties are the information envelope's own: it is* estimable*, because the plug-in $\hat{I}(X_t; X_{t-D})$ in §7 needs only pair counts, while the β-coefficient's supremum over the future σ-field is not estimable without a process model; it* carries rate semantics*, since total-variation dependence coefficients also contract under channels, but information naturally obeys the entropy bound $I(X; Y) \leq H(Y)$ that turns a bit budget into Corollary 3.2; and it obeys an* exact conditional chain rule*, giving additive multi-sensor budget decompositions via conditional mutual information (Theorem 3.4) that β does not. In short: mixing is the tighter analysis tool when a model is in hand; information is the operational one when only data or a bit budget is.*

Envelope	Bound on Δ_Y	Needs	Blind spot
master (TV)	$\Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$	posterior process	not estimable directly
mixing	$\Omega\,\beta(D)$	process model	future events beyond X_t (Prop. 5.2ii)
information	$\Omega\sqrt{I(X;Y)/2}$	pair samples	decision geometry (Prop. 4.1)
bit budget	$\Omega\sqrt{(b \ln 2)/2}$	channel rate only	everything but the rate

Table 2. The envelope family. The structure is a *branching*, not a hierarchy: both mixing and information relax the master TV envelope, in incomparable directions (Proposition 5.2), and the bit budget further relaxes the information branch via $I(X; Y) \leq H(Y)$: TV → { mixing; information → bit budget }. Each branch trades knowledge for tightness in its own currency; the battery of §7 measures the first three on real processes.

6 Exactness: a complete account for binary state, and its limits

When does the Bayes envelope admit a closed form, an exact *law* rather than an envelope? For binary state with a finite action set, the exact law is a sum of hinges; for general convex value functions, the *local contact order* of the value function at the prior determines the weak-correlation exponent (Theorem 6.3).

Theorem 6.1 (The hinge law, polyhedral case). *Let $X_t \in \{0, 1\}$ be stationary with $\pi = \mathbb{P}(X_t{=}1) \in (0, 1)$, let $Y = X_{t-D}$, and write $C = C(D) = \mathrm{Cov}(X_t, X_{t-D})$. Assume the action set is finite or, more generally, that the value function $\psi(p) = \sup_w [pR(w, 1) + (1 - p)R(w, 0)]$ is* polyhedral *(a maximum of finitely many affine functions of the belief), with kinks $p_1^* < \cdots < p_K^*$ and slope jumps*

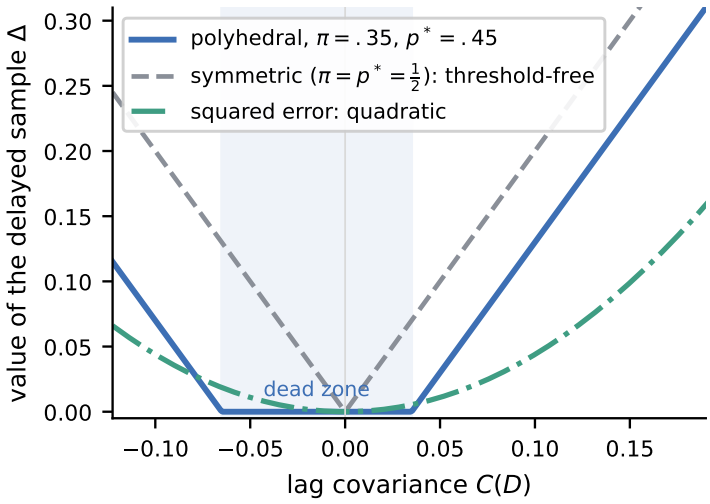


Fig. 1. The contact-order trichotomy (Theorem 6.3), drawn exactly. Polyhedral objective with π=.35, one kink at p^*=.45, Δs=2: sign-dependent hinges with dead-zone thresholds a^+=.035 (positive branch) and a^-=.065 (negative branch), so anticorrelation of the same magnitude can be worthless where positive correlation has value. The symmetric case is the threshold-free degenerate hinge; squared error responds quadratically with no dead zone. The axis spans exactly the feasible covariance range $[-\pi^2, \pi(1-\pi)]$; for stronger asymmetry (e.g. π=.2, p^*=.3) the negative-branch threshold exceeds the feasible range entirely and anticorrelated sensing is worthless at *every* achievable covariance.

$\Delta s_k > 0$. *Then, exactly,*

$$\Delta(C) = \begin{cases} \sum_k \Delta s_k \left(C - a_k^+\right)^+, & C \geq 0, \\ \sum_k \Delta s_k \left((-C) - a_k^-\right)^+, & C < 0, \end{cases} \qquad a_k^+ = \begin{cases} (p_k^* - \pi)\,\pi, & p_k^* \geq \pi, \\ (\pi - p_k^*)(1-\pi), & p_k^* < \pi, \end{cases}$$

$$a_k^- = \begin{cases} (p_k^* - \pi)(1-\pi), & p_k^* \geq \pi, \\ (\pi - p_k^*)\,\pi, & p_k^* < \pi. \end{cases}$$

Δ *is piecewise-linear and nondecreasing in* $|C|$ on each covariance-sign branch, *identically zero on the branch dead zone; unless* $\pi = \frac{1}{2}$ *or the value function has the corresponding reflection symmetry about* π *(kink locations* and *slope jumps), the two branches generally differ. Positive and negative correlation of the same magnitude can have different value, one of them zero.*

Proof. Write $\psi(p) = \psi(p_1^*) + s_0(p - p_1^*) + \sum_k \Delta s_k (p - p_k^*)^+$ (polyhedral by hypothesis). Observing Y produces the two-point posterior $p_1 = \pi + C/\pi$ (w.p. π) and $p_0 = \pi - C/(1-\pi)$ (w.p. $1-\pi$), whose mean is π; the affine part of ψ cancels in $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$, leaving $\Delta = \sum_k \Delta s_k \{\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*)^+\}$. Fix k. For $C \geq 0$ ($p_0 \leq \pi \leq p_1$): if $p_k^* \geq \pi$, $\mathbb{E}(p_Y - p_k^*)^+ = \pi(p_1 - p_k^*)^+ = (C - (p_k^* - \pi)\pi)^+$ and $(\pi - p_k^*)^+ = 0$; if $p_k^* < \pi$, the pointwise identity $(x)^+ - x = (-x)^+$ gives $\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*) = \mathbb{E}(p_k^* - p_Y)^+ = (1-\pi)(p_k^* - p_0)^+ = (C - (\pi - p_k^*)(1-\pi))^+$. For $C < 0$ the posteriors trade places ($p_1 < \pi < p_0$): if $p_k^* \geq \pi$, only p_0 can exceed the kink and $\mathbb{E}(p_Y - p_k^*)^+ = (1-\pi)(p_0 - p_k^*)^+ = ((-C) - (p_k^* - \pi)(1-\pi))^+$; if $p_k^* < \pi$, the same identity gives $\pi(p_k^* - p_1)^+ = ((-C) - (\pi - p_k^*)\pi)^+$. Summing over k gives the two branches. □

6.1 Beyond polyhedral objectives: contact-order response

The polyhedral hypothesis is not decoration. For a continuous action set ψ is a supremum of *infinitely* many affine functions and need not be piecewise-linear, and the hinge law fails, constructively:

Proposition 6.2 (Squared-error prediction: exactly quadratic). *For $w \in [0,1]$ and $R(w,x) = -(w-x)^2$ (oscillation $\Omega = 1$), $\psi(p) = -p(1-p)$ is smooth and strictly convex, and, exactly,*

$$\Delta(D) \;=\; \mathrm{Var}(p_Y) \;=\; \frac{C(D)^2}{\pi(1-\pi)}\,:$$

quadratic in the covariance, no hinge, no dead zone, for every π. More generally, whenever ψ is twice continuously differentiable near π with $\psi''(\pi) > 0$ (if the curvature vanishes the response is of higher order; Theorem 6.3),

$$\Delta(D) \;=\; \tfrac{1}{2}\,\psi''(\pi)\,\mathrm{Var}(p_Y) + o(C^2) \;=\; \frac{\psi''(\pi)}{2\pi(1-\pi)}\,C(D)^2 + o\big(C(D)^2\big).$$

Proof. The optimal action under belief p is $w = p$, so $\psi(p) = -p(1-p) = p^2 - p$; then $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi) = \mathbb{E}[p_Y^2] - \pi^2 = \mathrm{Var}(p_Y)$, exact because ψ is quadratic, and $\mathrm{Var}(p_Y) = \pi(C/\pi)^2 + (1-\pi)\big(C/(1-\pi)\big)^2 = C^2/\pi(1-\pi)$. The general statement is the second-order Taylor expansion of ψ at π against the mean-π posterior spread. □

Both regimes, and every intermediate one, are cases of a single statement:

Theorem 6.3 (Local response is set by the contact order). *Suppose that near π,*

$$\psi(\pi + h) \;=\; \psi(\pi) + ah + c_+\,h_+^q + c_-\,(-h)_+^q + o(|h|^q)$$

for some $q \geq 1$ and $c_+, c_- \geq 0$ not both zero. Then, as $C \downarrow 0$,

$$\Delta(C) \;=\; C^q\Big[c_+\,\pi^{1-q} + c_-\,(1-\pi)^{1-q}\Big] + o(C^q),$$

and on the negative branch ($C \uparrow 0$) the same with $|C|^q$ and the weights $\pi \leftrightarrow (1-\pi)$ interchanged. If instead ψ is affine on a neighbourhood of π, then $\Delta(C) = 0$ for all $|C|$ small enough that both posteriors remain in that neighbourhood. This is the exact dead zone.

Proof. The posteriors are $p_1 = \pi + C/\pi$ (w.p. π) and $p_0 = \pi - C/(1-\pi)$ (w.p. $1-\pi$); their mean is π, so the affine part of the expansion cancels in $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$. For $C > 0$ the h_+^q term contributes $\pi\,c_+(C/\pi)^q = c_+C^q\pi^{1-q}$ through p_1 and the $(-h)_+^q$ term contributes $(1-\pi)\,c_-(C/(1-\pi))^q = c_-C^q(1-\pi)^{1-q}$ through p_0; both posterior deviations are $\Theta(C)$, so the $o(|h|^q)$ remainder is $o(C^q)$. For $C < 0$ the posteriors trade sides of π, exchanging the weights. The affine case is immediate. □

Remark 6.4 (The trichotomy, as leading cases). *The three named regimes (dead zone, linear at a kink, quadratic under smooth curvature) are precisely De Lara and Gossner's small-information trichotomy ([3], Thms. 3, 5 and 7, illustrated in their Prop. 10), and their exact zero-value characterization (their Prop. 2, the confidence-set condition) is the posterior-space form of our dead zone. Theorem 6.3 adds what their bounds deliberately do not:* exact leading coefficients *(their two-sided constants "depend on the global behavior of the value function, and hence cannot be inferred from local properties only," whereas in the delayed-binary geometry the coefficient $C^q[c_+\pi^{1-q} + c_-(1-\pi)^{1-q}]$ is explicit), together with general fractional contact orders $q \in (1,2)$, realizable by valid suprema of affine rewards, and the lag-covariance parametrization with its sign asymmetry. Each case fixes how weak dependence buys value:* prior on a kink ⇒ *linear response $\Delta \propto |C|$;* prior interior to a flat piece ⇒ *dead zone, $\Delta = 0$ until a threshold;* smooth curvature ⇒ *quadratic response $\Delta \propto C^2$. The companion's law is the first case. Symmetry puts the prior on the decision kink, the polyhedral*

configuration yielding threshold-free linear response. The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [18, 19], whose smooth setting is our third case (zero marginal value at zero information, but not identically zero). And the sign-dependence of Theorem 6.1 means even the direction *of correlation matters away from symmetry: with $\pi = 0.2$ and one kink at $p^* = 0.3$, covariance $+0.03$ clears its threshold $(0.1)(0.2) = 0.02$ and has value, while -0.03 sits inside its threshold $(0.1)(0.8) = 0.08$ and is worthless.*

Corollary 6.5 (The companion law, and why it is threshold-free). *For the symmetric chain ($\pi = \frac{1}{2}$) with the 0/1 prediction objective (K=1, $p^* = \frac{1}{2}$, $\Delta s = 2$): $a^{\pm} = 0$ and $|C(D)| = \frac{1}{4}|r(D)|$, so $\Delta(D) = 2 \cdot \frac{1}{4}|r(D)| = \frac{1}{2}|r(D)|$, the exact law of [1]. Symmetry places the prior* on *the kink, the polyhedral configuration with no dead zone; the linear-in-r law is the degenerate hinge.*

Remark 6.6 (The dead zone is a prediction, not a curiosity). *For $\pi \neq p^*$ the hinge says weakly-correlated delayed feedback is* worthless, *not slightly valuable, until $|C(D)|$ crosses its branch threshold $a^{\pm}$: below threshold, both posteriors fall on one side of the decision kink and the optimal action never changes. Operationally: a rare-event admission controller (π small, p^* moderate) gains* nothing *from telemetry until the covariance clears the* branch *threshold: $C(D) > (p^* - \pi)\pi$ for positive correlation, but $-C(D) > (p^* - \pi)(1 - \pi)$ for negative. For rare events (π small) the negative-branch threshold is larger by the factor $(1 - \pi)/\pi$, so anticorrelated telemetry must be far stronger before it is worth anything. This is an exact, testable, sign-dependent cutoff for when to stop paying for freshness at all. None of the envelopes of §5 see this: the dead zone is a fact about geometry (posteriors vs. kink), which is precisely what scalar information summaries discard (Proposition 4.1).*

Remark 6.7 (Beyond two states: a layer-cake decomposition, and the boundary). *For capacities $a_t \in \{0, \ldots, C\}$ with monotone threshold objectives, the objective decomposes over layer indicators $x_t^u = \mathbf{1}[a_t \geq u]$, each layer a binary experiment governed by its hinge (sketch in Appendix A). Against a comonotone static budget at matched over-admission, a layer with $r_u \geq 0$ contributes $\pi_u r_u(D)$, because at matched surplus each layer's kink sits at its own prior and no layer has a dead zone. This is a matched-surplus benchmark gain, not the Bayes envelope of* (1). *The companion leaves integer capacity open: the per-layer decisions must stay nested to form one admission window, and without a monotonicity condition the layerwise sum is only an upper bound on the jointly feasible advantage. We do not close that gap here. Effective one-dimensionality of the posterior family (a scalar belief per layer, moved along a line by the observation) is a principal tractable regime in which scalar closed forms arise; the two-bit construction of Proposition 4.1 is the minimal experiment whose posterior geometry is irreducibly two-dimensional, and there the value ceases to be a function of any scalar summary of dependence, correlation or information alike.*

7 The law and its envelopes across domains

The companion paper had room for one paragraph of this study; the full protocol and results are reported here. The question has two halves. First, *single-sample sufficiency*: on real binary state processes, does history beyond the single delayed sample add decision value beyond the *exact empirical single-sample Bayes value*, computed from the pooled 2×2 lag table with no symmetry assumption, so that Theorem 6.1's imbalance effects (dead zones included) are carried automatically? Second, *the envelope*: does the measured value respect, and how tightly does it approach, the predictive-information bound of Theorem 3.1?

One caution frames everything that follows. Every battery quantity is a property of the empirical mixture a series samples. A history gain pooled over a whole record can come from identifying the regime as well as from prediction within one (Lemma 3.5), and every series here rejects a homogeneous lag-1 law across calendar blocks, so the section reports the pooled results and the block results side by side.

7.1 Protocol

Throughout this section, "registered" means fixed in the protocol code before the final analysis runs (the introduction calls this protocol-fixed). The battery was not preregistered with a third party, and its protocol was revised during the v3 and v4 development runs, whose record is part of the released artifact.

Stationarity. The envelopes of §3 need no stationarity. The covariance parametrization of §6, the fitted homogeneous Markov null, and the empirical dead-zone threshold all assume stationary marginals, and the GOES and seismic series are plausibly piecewise stationary at best. Pooling a long nonstationary series can make history valuable because it identifies the regime or the time, not because of stationary higher-order dependence. We therefore read every battery quantity, including the seismic certificate, as a property of the empirical mixture the series samples, not as a parameter of a stationary process. Lemma 3.5 makes the distinction exact: a history gain pooled over the whole record bounds what history reveals about the regime plus what it predicts within one, and a gain measured inside a calendar block bounds the second term alone; the regime analysis at the end of this section measures both.

Artifact. An anonymized artifact is available for review at https://osf.io/273mh/overview?view_only=0b9598b1fdd84fd88688089e3408cfd1 and will be archived on Zenodo on acceptance: the code, the derived binary series, every result file, the Lean proof-check logs and MATLAB symbolic-verification logs, the seeds of every resampling step, a verification record for every reference, and a notebook that rebuilds every table and figure from the result files and checks each printed certificate (all 290 checks pass).

Use of generative AI. An AI assistant (Claude, Anthropic) was used throughout as a research assistant, in the following specific ways. It wrote, under the author's direction, the battery drivers (versions 3 to 5c), the stationarity and BGP analysis scripts, and the figure script, and it ran them on the author's workstation. It drafted the protocol for the BGP series and the stationarity check, which the author approved before any data were fetched. It reviewed the proofs by independent re-derivation, audited every reported number against the result files, and searched the literature for prior statements of the results (the attribution of Theorems 3.1 and 3.4 came from that search). It wrote the Lean scripts that machine-check the finite and algebraic results and the MATLAB scripts that symbolically verify the analytic results, and it drafted and copy-edited text. The author chose the questions, the theory and the battery design, checked the outputs, and is responsible for every claim.

States. Binary regime series from four physical domains, one computer system, and controls: acoustic source azimuth and voice activity (LOCATA; azimuth at 120 Hz, voice activity at 100 Hz); sperm-whale coda type (Dominica corpus, per-exchange); GOES space-weather series (X-ray flux, magnetometer, proton flux; 1–5-minute cadence); seismic ground motion at station IU.ANMO (1 Hz over 3.5 hours, from 30 minutes before the origin of the 2015 Mw 8.3 Illapel, Chile earthquake, so the record holds the teleseismic arrivals and their coda); BGP control-plane activity at RIPE RIS collector rrc06 (1-minute, 1 to 7 September 2026, state 1 when the minute's count of announced or withdrawn prefixes exceeds the week's median, a rule registered before the data were fetched); a synthetic two-state Markov chain (flip probability 0.02) as positive reference; an i.i.d. fair-bit series as null. Each series is binarized by the domain's registered thresholding (the companion study's preparation scripts, unchanged).

The two estimands. For each lag D on a per-domain grid: (i) the *exact empirical single-sample Bayes value* $\widehat{V}_1(D) = \sum_y \hat{w}_y \max(\hat{p}_y, 1-\hat{p}_y) - \max(\hat{\pi}, 1-\hat{\pi})$, the plug-in of (1) for $Y = X_{t-D}$ computed from the pooled lag table. For a balanced series this is $\frac{1}{2}|r(D)|$, and for an imbalanced one it carries Theorem 6.1's thresholds automatically (the per-domain class proportion $\hat{\pi}$ is reported

alongside); (ii) the value of the best *delayed-history* policy: a k=6-lag lookup predictor trained and evaluated by blocked cross-validation (5 folds, contiguous blocks), reported as accuracy above the *best static (majority) policy* $\max(\hat{\pi}, 1-\hat{\pi})$, the correct static benchmark; accuracy above $\frac{1}{2}$ overstates value whenever $\hat{\pi} \neq \frac{1}{2}$. The *history gain* is the difference (ii)−(i).

Scope of the estimand. Estimand (ii) is the value of one *registered policy class* $\mathcal{P}$ (six-lag lookup), i.e. $\widehat{\Delta}_{\mathcal{P},W} = \sup_{\pi\in\mathcal{P}} V(\pi;W) - V_{\text{stat}}$, *not* the Bayes history value Δ_W of Theorem 3.4. The asymmetry matters. Since $V(\pi;W) \le V_W^\star$ for every π, any demonstrated improvement over the exact single-sample value lower-bounds $\Delta_W - \Delta_Z$, so *positive* certificates transfer upward and the conditional-information witnesses below are sound. *Negative* findings bound only the class: where the lookup policy finds nothing, some richer policy class could still extract history value, so verdicts of that kind are stated as "no additional value detected by the registered six-lag lookup policy," never as unrestricted single-sample sufficiency.

Surrogate calibration. The null for "history adds nothing beyond the law" is a population of Markov surrogates per domain, each matched to the real series' lag-1 transition probabilities and evaluated by the identical pipeline, with $M = 299$ surrogates. We name the procedure precisely: because the surrogates are generated from *estimated* lag-1 transition probabilities, this is a *parametric-bootstrap test under a fitted Markov null* (not an exact randomization test for the broad null "history adds nothing"), with $p = (1 + \#\{T_j \ge T_{\text{obs}}\})/(M+1)$ (smallest reportable value $1/300 \approx 0.0033$, written $p < .004$, never .00), together with the companion study's practical floor of 0.03 on the effect. Because each surrogate is Markov, its true history gain is zero and its own exact $\widehat{V}_1$ is computable from its lag table, so the M surrogate gain estimates sample the pipeline's estimation error E *under the fitted Markov model*. The two tails play different roles: a lower certificate needs the *over*-estimation tail, $\varepsilon_{\text{over}} = q_{0.95}(E)$, giving $L = \hat{g} - \varepsilon_{\text{over}}$; an upper (equivalence) bound needs the *under*-estimation tail, $\varepsilon_{\text{under}} = -q_{0.05}(E)$, giving $U_2 = \hat{g} + \varepsilon_{\text{under}}$; the two coincide only under a symmetry condition, so both are computed and reported separately. Both are *model-calibrated* (not distribution-free) bounds. Complementing them, the analysis reports a *dependence-aware* moving-block bootstrap on held-out paired loss differences, comparing the registered k=6 policy against the pooled-*exact* single-sample policy (in-sample fit favouring the single-sample arm, hence conservative for the incremental value), blocks resampled within runs at the selected lag D^*. Because D^* is an argmax over the lag grid that the bootstrap treats as fixed, the i.i.d. null's paired interval calibrates the *selection* bias not covered (its upper end ≈ 0.02); paired lower bounds are claimed only above that floor. The selection-aware refinement has been executed: repeating the entire lag selection inside each block-bootstrap replicate (blocks resampled with shared starts across all registered lags, the maximizing lag re-selected per replicate, the one-sided bound taken from the resulting max-statistic distribution). The i.i.d. null quantifies the percentile-of-max procedure's residual optimism at +0.008, so certificates are claimed only where the parametric-bootstrap detection test also fires, which excludes the noisy short-series maxima the winner's curse can manufacture (whale, protons). Validity checks pass: the Markov reference's interval straddles zero (no phantom conditional information) and azimuth's is slightly negative (the overfitting cost of the richer class). In plain terms: choosing the best of many lags and then quoting its interval overstates the gain, because the best lag is partly the luckiest. The fixed-lag bound $L^{\text{blk}}_{D^*}$ ignores that choice; the selection-aware bound $L^{\text{blk}}_{\max}$ repeats the choice inside every bootstrap replicate, so it bounds the gain of whichever lag wins, and the i.i.d. null, which has no true gain at any lag, shows how much optimism remains (+0.008).

The practical floor. Every verdict compares a bound with a practical floor of 0.03. On the 0/1 prediction objective ($\Omega = 1$) the gain is a difference in the probability of a correct decision, so the floor is three percentage points of correct decisions per decision. It was fixed with the equivalence

rule before the final runs, and it sits above the procedure's own false-positive floor, the upper end of the i.i.d. null's paired interval (≈ 0.02), and above the selection-aware residual optimism (+0.008). It does not come from a cost model, so we report how the verdicts move with it. Seismic is "history adds" at floors of 0.02, 0.03 and 0.05. BGP is "history adds" at 0.02 and 0.03 and only detected at 0.05, where its lower bounds (+.044, +.047) fall short. Magnetometer becomes "history adds" at 0.02; at 0.05 the whale codas and protons become equivalent (Table 3).

Series	floor 0.02	floor 0.03 (used)	floor 0.05
LOCATA azimuth	not established	equivalent	equivalent
Sperm-whale codas	inconclusive	inconclusive	equivalent
GOES protons	not established	not established	equivalent
GOES X-ray flux	detected (exploratory)	detected, below floor	detected, below floor
GOES magnetometer	history adds	detected (exploratory)	detected, below floor
Seismic ANMO	history adds	history adds	history adds
BGP rrc06	history adds	history adds	detected (exploratory)

Table 3. Verdicts at three practical floors, from the same result files. Series not listed (two-state Markov, LOCATA VAD) are equivalent at every floor, and the i.i.d. null is the selection control throughout.

Non-rejection is not exactness: where the test does not fire, we report the model-calibrated two-tail upper bound $U_2 = \text{gain}_{\max} + \varepsilon_{\text{under}}$ and claim "no detectable excess (equivalent within the margin)" only when $U_2 < 0.03$.

The two information measurements. For each domain and lag, the plug-in $\widehat{I}(X_t; X_{t-D})$ from pooled pair counts (2×2; first-order plug-in bias $O(1/N)$, negligible at the pooled sample sizes reported) and the envelope $\sqrt{\widehat{I}/2}$. Their registered uses follow the two theorems, matched to the observable each governs. *Single sample* (Theorem 3.1 at $Y = X_{t-D}$): the exact $\widehat{V}_1$ must respect $\sqrt{\widehat{I}/2}$, a population identity whose empirical check is a consistency diagnostic. *History* (Theorem 3.4): the history policy observes the *window*, so the single-lag envelope does not bound it. A history value above $\sqrt{\widehat{I}_1/2}$ is not a violation but exactly what genuine higher-order dependence should produce. The history gain over $\widehat{V}_1$ is a registered-policy gain that *lower-bounds* the Bayes increment $\Delta_W - \Delta_Z$, and its model-calibrated lower bound therefore also bounds *conditional* predictive information:

$$I(X_t;\ \text{history} \mid X_{t-D}) \ \geq\ 2\left[(\text{gain}_{\max} - \varepsilon_{\text{over}})^+\right]^2,$$

never through the raw point estimate, whose positive noise would inflate the certificate. The total-information witness $I(X_t; \text{window}) \geq 2[(\widehat{\Delta}_{\text{hist}} - \varepsilon_{\text{over}})^+]^2$ is retained separately, with the caveat that its estimator differs from the incremental-gain estimator and in principle requires its own overestimation calibration; the gain-error quantile is reused as an approximation and labelled as such.

7.2 Results

Table 4 reports the final analysis from four analysis runs of the released artifact. Run v5 gives the gains, p and calibration tails, v5b the fixed-D^* bounds and v5c the selection-aware bounds. Run v4 gives the single-sample quantities $\widehat{V}_1$ and $\widehat{I}$, the tightness ratios, Fig. 2 and the total-information witness. The BGP series and the calendar-block analysis come from separate runs of the same code, registered in the artifact before the BGP data were fetched. The v3 run and the protocol corrections are preserved in the artifact. Per domain: the class proportion $\hat{\pi}$; the maximum history-class gain over the exact single-sample Bayes value $\widehat{V}_1$; the parametric-bootstrap p under the fitted

Markov null (M=299, floor 1/300, written $<$ .004); the two calibration tails $\varepsilon_{\text{over}} = q_{0.95}(E)$ and $\varepsilon_{\text{under}} = -q_{0.05}(E)$; the two-tail equivalence bound $U_2 = \text{gain}_{\text{max}} + \varepsilon_{\text{under}}$ (margin 0.03); and the dependence-aware paired interval, a moving-block bootstrap (blocks within runs, B=2000) of the per-position loss difference between the registered k=6 policy and the pooled-*exact* single-sample policy at the selected lag D^*, the exact-arm fit favouring the single-sample side so the interval is conservative for the incremental value.

Domain	$\hat{\pi}$	gain	p	ε_{ov}	ε_{un}	U_2	$L^{\text{blk}}_{D^*}$ / $L^{\text{blk}}_{\text{max}}$	verdict
Two-state Markov	.470	+.000	.78	.006	.000	.000	−.003 / +.000	equivalent
i.i.d. null	.497	+.012	.023	.011	−.001	.011	+.005 / +.008	selection control
LOCATA azimuth	.494	+.008	.063	.009	.020	.027	−.003 / −.001	equivalent
LOCATA VAD	.636	−.000	.82	.052	.000	.000	.000 / .000	equivalent
Sperm-whale codas	.483	+.046	.997	.117	−.063	−.016	+.010 / +.040	inconclusive‡
GOES protons	.489	+.029	.11	.033	.003	.032	+.002 / +.020	not established
GOES X-ray flux	.421	+.028	$<$ .004	.008	.001	.028	+.011 / +.017	detected, below floor
GOES magnetometer	.456	+.033	$<$ .004	.010	.001	.033	+.022 / +.026	detected (exploratory)
Seismic ANMO	.508	+.158	$<$ .004	.014	.000	.158	**+.146 / +.145**	history adds
BGP rrc06	.500	+.053	$<$ .004	.015	−.002	.051	+.044 / +.047	history adds

Table 4. Battery v5 (final analysis). Verdicts are statements about the *registered six-lag lookup class*. "Detected" means the parametric-bootstrap p is at its floor ($<$ .004); no series has p between .004 and .023, so any level in that range gives the same verdicts. This is a conservative analysis rule adopted during the final analysis, not a preregistered significance level. The verdicts, applied in this order: "history adds" = detected, with both moving-block lower bounds above the practical floor 0.03 (a certificate is issued); "detected (exploratory)" = detected, point gain above the floor, lower bounds below it; "detected, below floor" = detected, point gain at or below the floor; "equivalent" = not detected and $U_2 < 0.03$; "not established" = not detected and $U_2 \geq 0.03$; "inconclusive" = the calibrations disagree (footnote). Detection takes precedence, so X-ray, which also has $U_2 < 0.03$, is reported as detected. "Exploratory" refers to the size of the gain, not to the calendar-block analysis. The i.i.d. null (p = .023) is the selection control. $L^{\text{blk}}_{D^*}$ and $L^{\text{blk}}_{\text{max}}$ are the one-sided 95% moving-block lower bounds at the pre-selected lag and with lag selection repeated inside each replicate (the latter bounds the *maximal* gain over the grid). ‡Inconclusive: the calibrations disagree. The fitted-Markov calibration finds the estimator biased upward ($U_2 < 0$), while both moving-block lower bounds are positive and the selection-aware one (+.040) exceeds the practical floor, and the detection test does not fire on this short series (p = .997; N = 4,698 pairs, the second-shortest, Table 6). We claim neither equivalence nor a certificate. The i.i.d. null's *positive* paired interval is the empirical lag-selection bias (its upper end, ≈ 0.02, is the claim floor for paired bounds, since D^* is an argmax the bootstrap treats as fixed). Bold: the one interval that dwarfs every calibration concern. Sample sizes, lag grids, selected lags and block lengths are in Table 6.

Four readings, in increasing order of consequence.

The protocol corrections changed the conclusions. Under the original chance baseline, four series appeared to carry history value beyond the single sample (gains 0.100–0.367). Under the majority baseline and exact $\widehat{V}_1$: VAD's gain vanished entirely (+0.100 → −0.000; $\hat{\pi}$ = .636, so the whole effect was the majority-class head start), X-ray shrank 0.185 → 0.028, magnetometer 0.147 → 0.033, and only seismic survived at strength (0.367 → 0.158). The two-tail calibration then *strengthened* the equivalence side: $\varepsilon_{\text{under}}$ is near zero or negative on most series, so VAD joins the class-equivalent block. On the whale codas the two calibrations point in opposite directions, and that series is reported as inconclusive. Sub-unity ratios of the envelope to the exact single-sample value disappeared with the imbalance correction. On the series with $|\hat{\pi} - \frac{1}{2}| < .035$ the median ratio $\sqrt{\hat{I}/2}/\widehat{V}_1$ runs from 1.01 (BGP) to 1.35 on the six such series, above one as Theorem 3.1 requires.

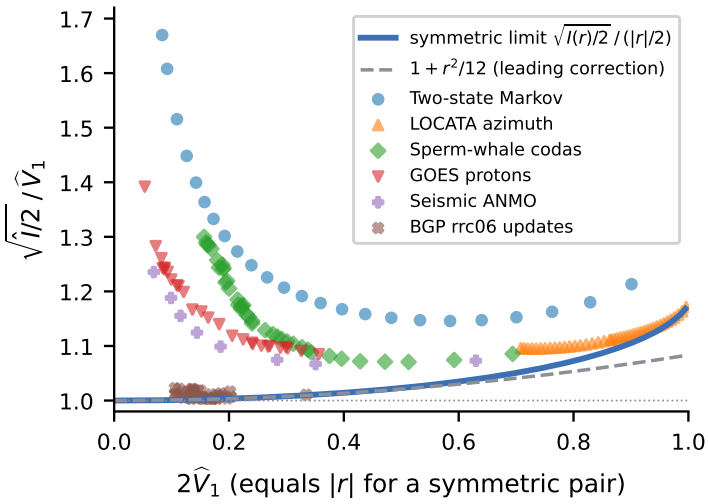


Fig. 2. Tightness of the information envelope on approximately balanced binary pairs (Proposition 3.6). Points: per-lag ratio $\sqrt{\hat{I}/2}\,/\,\widehat{V}_1$ of the information envelope to the exact single-sample value, both computed from the same pooled lag table, for the series with $|\hat{\pi} - \frac{1}{2}| < .035$ (lags with $\widehat{V}_1 > .025$; the i.i.d. null has none). Theorem 3.1 applied to that table forces every point to at least one. The horizontal axis is $2\widehat{V}_1$, which equals $|r|$ for a symmetric pair, so the curve is the symmetric limit $\sqrt{I(r)/2}\,/\,(|r|/2)$ and the dashes its leading correction $1 + r^2/12$. Points above the curve at small $\widehat{V}_1$ are the mildly imbalanced series, whose single-sample value is pulled toward Theorem 6.1's threshold (Fig. 1). A consistency check, not an independent test.

This is a consistency check, not an independent test, since $\hat{I}$ and $\widehat{V}_1$ come from the same pooled lag table. In Fig. 2, the points above the symmetric limit at small $\widehat{V}_1$ are the mildly imbalanced series.

The controls behave as designed. The i.i.d. null has no true history value, and its positive bounds (+.005 fixed-lag, +.008 selection-aware) are the procedure's false-positive floor, which every claimed gain must clear. The two-state Markov reference, whose single delayed sample is sufficient by construction, comes out equivalent. VAD's fixed-lag entry is exactly zero for a structural reason: at its selected lag of one step the six-lag lookup policy and the exact single-sample policy reduce to the same majority rule on every fold.

Two series exhibit material history gains. Seismic and BGP are detected with both moving-block lower bounds above the practical floor. X-ray and magnetometer show smaller statistically detected gains near the registered practical threshold (point gains .028 and .033 against .03, lower bounds below it); on the remaining series *no additional value was detected by the registered six-lag lookup policy*, with class equivalence established where $U_2 < 0.03$ and withheld where it is not (protons, $U_2 = .032$). None of these negative findings asserts unrestricted single-sample sufficiency: a richer policy class could in principle extract value the lookup class does not (§ protocol, scope of the estimand).

An empirical dead zone. The imbalanced GOES series exhibit Theorem 6.1's geometry at the plug-in level: X-ray ($\hat{\pi} = .421$, kink at $\frac{1}{2}$, positive-branch threshold $a^+ \approx 0.033$) has $\widehat{V}_1$ exactly zero across its moderate-covariance lags (the majority policy is unbeatable by any single delayed sample), while its plug-in information remains positive: *information without value*, the phenomenon

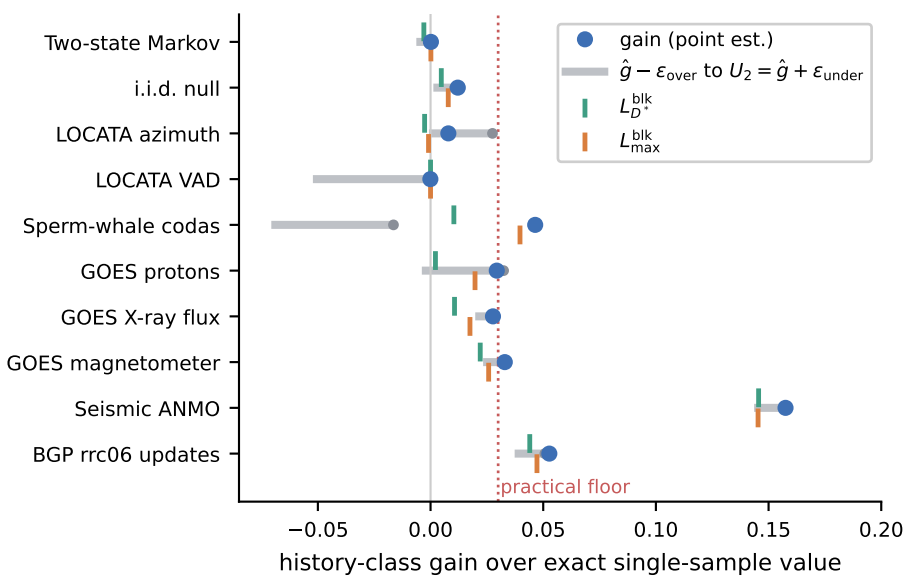


Fig. 3. The battery at a glance. Dots: maximum history-class gain over the exact single-sample value; grey bars: the model-calibrated interval $[\hat{g} - \varepsilon_{\text{over}}, \hat{g} + \varepsilon_{\text{under}}]$; ticks: the one-sided 95% moving-block bounds ($B = 2000$ replicates) at fixed D^* (upper, teal) and with in-replicate lag selection (lower, orange); solid vertical line: zero; dotted vertical line: the registered practical floor 0.03. Seismic and BGP separate from both zero and the floor; the i.i.d. null's positive selection-aware tick is the empirical winner's-curse floor; whale's wide bar is the short-series noise the verdict language respects. Where $\varepsilon_{\text{under}} < 0$ (whale, i.i.d.) the bias-corrected upper endpoint U_2 sits *below* the point estimate, because the estimator is upward-biased there, and the bound says so.

of Proposition 4.1, on a physical series. (Population uncertainty for the dead-zone boundary itself is future work; we label this an *empirical* dead zone.)

The pooled conditional lower bounds. By Theorem 3.4, each detected history gain converts into a lower bound on conditional predictive information over its whole record. The *primary* result is dependence-aware and empirically selection-calibrated (the i.i.d. null measures the procedure's residual optimism; we give no formal coverage argument for the maximum), with the correctly-stated target: inference on the per-replicate maximum bounds the *maximal* gain over the registered lag grid, so the certificate reads $\max_{D \in \mathcal{D}} I(X_t; \text{history}_D \mid X_{t-D}) \geq 2(0.1453)^2 \approx 0.042$ nats (0.061 bits, about 6% of the entropy of a fair binary state, so a predictor that also reads the history can lower its expected log-loss by at least that much). In words, *there exists a registered lag at which the conditional predictive information is at least* 0.042 *nats*. (A claim at the specific D^* would need a simultaneous band or sample splitting; the fixed-D^* interval below is the conditional-on-D^* statement.) Two further routes corroborate to within a few percent: the fixed-D^* moving-block interval ($[+.146, +.168]$, ≈ 0.042) and the fitted-Markov surrogate calibration ($2(0.1575 - 0.0139)^2 \approx 0.041$), alongside the *approximate* total-information witness $I(X_t; \text{window}) \gtrsim 0.176$ nats (0.254 bits) (exploratory: its estimator reuses the gain-error calibration, as noted in the protocol). The same route gives BGP $2(0.0472)^2 \approx 0.0045$ nats (fixed-D^* bound +.044; surrogate route $2(0.0527 - 0.0154)^2 \approx 0.0028$). Magnetometer's selection-aware bound (+.026, certificate ≈ 0.001 nats) is small and reported as exploratory. The instrument reads in both directions: class equivalence or bounded excess on

most series; a dependence-aware conditional-information lower bound exactly where the history structure is found.

Regimes. The stationarity check was registered with the BGP series before its data were fetched: each of the five series with a detection splits into three equal contiguous calendar blocks, and the seismic certificate counts as block-robust only if its selection-aware bound clears the practical floor in all three. *It does not.* Block 1, which is mostly the 30 minutes before the earthquake's waves arrive, gives +.014 ($p = .45$, no detection), while blocks 2 and 3 give +.219 and +.165, certificates of 0.096 and 0.054 nats, both larger than the pooled 0.042 (Table 5). BGP shows the same shape: a detection below the floor in block 1, history adds in block 2 (certificate 0.0062 nats), and no material gain was detected by the registered lookup policy in block 3, while its class proportion drifts from .65 to .32 under the single weekly threshold. The GOES detections do not reach the bootstrap floor in any block, and a chi-square test rejects equal lag-1 transition matrices across blocks for every series ($p \le 10^{-4}$). By Lemma 3.5, the pooled certificates bound regime identification plus within-regime prediction, and the block certificates isolate the second. On both the earthquake record and the BGP week, the registered lookup policy detects material within-block value in selected blocks and no material value in the remaining blocks; a richer policy class could find value the lookup class does not. For a controller this is the operational reading. The value of delayed history is a property of the regime, and the instrument says where it pays. The registered robustness test failed and is reported as failed; the block results are descriptive.

Series	block	$\hat{\pi}$	gain	p	$L^{\mathrm{blk}}_{\max}$	verdict
Seismic ANMO	1	.513	+.012	.45	+.014	equivalent
	2	.501	+.237	$< .004$	**+.219**	adds (0.096 nats)
	3	.514	+.181	$< .004$	**+.165**	adds (0.054 nats)
BGP rrc06	1	.649	+.010	$< .004$	+.001	below floor
	2	.527	+.062	$< .004$	+.056	adds (0.006 nats)
	3	.323	+.003	.20	−.003	equivalent
GOES X-ray	1/2/3	.37/.42/.47	+.020/+.036/+.006	.017/.007/.66	+.013/+.017/+.005	none detected
GOES magn.	1/2/3	.47/.45/.45	+.030/+.024/+.022	.040/.057/.043	+.019/+.028/+.001	none detected
GOES protons	1/2/3	.50/.48/.48	+.017/+.055/+.040	.76/.12/.33	+.003/+.026/+.027	none detected

Table 5. Registered calendar-block analysis (three equal contiguous blocks per series; same pipeline and verdict rules as Table 4). The pre-stated seismic test (selection-aware bound above 0.03 in every block) fails in block 1. "Adds" means history adds, with its certificate. Both history-adds series carry their value in one or two regimes, and the block certificates exceed the pooled ones where history pays. Homogeneity of the lag-1 transition matrices is rejected for every series (chi-square, df = 4, $p \le 10^{-4}$).

8 Related work

Value of information. The Bayes envelope is the classical object of statistical decision theory [2, 4], and its convex geometry is charted territory: De Lara and Gossner [3] study the value function as a support function, characterize *exactly* when information has zero value (their Prop. 2), and prove the small-information trichotomy our Theorem 6.3 generalizes (their Thms. 3, 5 and 7; see Remark 6.4 for the precise delta). Our Lipschitz lemma instantiates their payoffs–beliefs duality; the sum-over-kinks skeleton of Theorem 6.1 descends from the mixture-of-primitives representations of statistical information for binary experiments [22, 23], and what is new there is the exact lag-covariance parametrization with explicit sign-asymmetric thresholds, a form we have not found in that literature, nor in the nonconcavity strand [18, 19, 21, 24]; the comparison-of-experiments order behind Lemma 3.3 is Blackwell's [17]; and the Radner–Stiglitz nonconcavity [18, 19] anticipates

the zero-marginal-value onset in smooth settings; the exact dead zone is its polyhedral counterpart.

Minimum excess risk. The envelope inequalities of §3 are published results of the minimum-excess-risk literature: the unconditional bound in [11–13], the conditional form in [14, 15], the proof engine in [16]; Fano-type families bounding Bayes *error* by information are the classical ancestors. That literature also contains the exact logarithmic anchor (MER under log loss *equals* conditional information; [14], Lem. 2), which joins the Kelly–Barron–Cover bracket below. What we have not found there, in the economics of information or elsewhere, is the value-of-delayed-observation framing, the exact covariance-parametrized laws of §6, the envelope *anchored to an exact law* (sharp to $O(r^3)$), its ordering against mixing envelopes with both separations constructed, and its inversion into certified conditional-information witnesses.

The logarithmic exception. For log-utility gambling the value of side information *equals* mutual information: Kelly [5] for the growth rate, Barron and Cover [6] for the general bound $\Delta \le I$ with equality characterizations. Log utility is unbounded, escaping our Ω-bounded class; the pair of results brackets the landscape. Unbounded log utility buys equality, bounded utility only an envelope, and Proposition 4.1 shows the gap is essential, not technical.

Age of information, and its critics. AoI [7] treats freshness as the object, with sampling-policy results such as [8] optimizing age processes. That age alone lacks decision significance has already been argued *within* the field: age-of-incorrect-information replaces the clock with an error-aware penalty [25], explicit AoI-vs-VoI scheduling comparisons exist for networked control [26], and value-of-information metrics for feedback control have been quantified directly, including an explicit derivation of the value–age relation [20, 27]. Our claim is accordingly not that age lacks decision semantics (that point is established) but more specific: for exogenous one-step bounded-reward decisions the exact semantic object is the posterior Bayes envelope, which admits a universal mutual-information converse, an explicit binary geometry (hinges, dead zones, quadratic regimes), and an operational inversion into predictive-information witnesses.

Delayed information in control. Delayed-sharing information structures go back to Witsenhausen [9] and are systematized by the common-information approach [10]; remote estimation studies the estimation face of the same trade [8]. Our setting is the tractable exogenous corner of that world (the action does not affect the state), which is exactly what makes closed forms and universal envelopes available.

Predictive information. Bialek, Nemenman and Tishby [28] study $I(\text{past};\text{future})$ as a measure of process complexity. Theorems 3.1 and 6.1 are, in that vocabulary, statements about what predictive information can and cannot buy a bounded-utility decision maker, with the battery's witness bound $I \ge 2[(\widehat{\Delta} - \varepsilon)^+]^2$ giving a one-sided lower bound on predictive information from policy value alone.

9 Conclusion

The companion paper measured one exact law on one system. This paper locates that law inside a unified theory of exogenous one-step bounded-reward decisions under delay: value of delayed information is the Bayes envelope; the envelope is bounded by predictive information (universally), by mixing (given a model), and by the entropy of the observation (given nothing); for binary state it is a sign-dependent sum of hinges under polyhedral objectives (a smooth value function giving quadratic weak-correlation response instead), and exactly nothing (inside the dead zone) when the prior sits off a kink and the coupling is weak. Beyond one effective dimension, no scalar summary of dependence determines value at all. Each claim is falsifiable, and the ten-series battery exercises the operational claims: the protocol-fixed policy shows no detectable history excess where the model-calibrated equivalence criterion is met; the information envelope holds on every real pair table as a consistency check; policy gains yield dependence-aware lower bounds on conditional

predictive information; and calendar blocks identify where those gains occur. The instrument reads in both directions; that, more than any single inequality, is what we hope survives.

A Companion results restated

No claim in this paper depends on the companion. Every result used from it is stated and proved below: the Lipschitz lemma (Lemma 2.1), the binary chain's mixing coefficient and exact law, and the layer-cake sketch. Theorems 3.1, 3.4 and 6.1 are this paper's own statements; the first two restate published inequalities (§3) and are proved in full in the text.

Mixing coefficient. Throughout, $\beta(D) = \mathbb{E}[\sup_{B\in\mathcal{F}_{\geq t}} |\mathbb{P}(B \mid \mathcal{F}_{\leq t-D}) - \mathbb{P}(B)|]$ (event-supremum convention; a lower bound for the absolute-regularity coefficient, so the vacuousness results transfer conservatively).

Lipschitz value (Lemma 2.1). Let $\Omega \geq \sup_x R(w,x) - \inf_x R(w,x)$ for every action w. Because $\sum_x(\mu(x) - \mu'(x)) = 0$, subtracting the midrange of $R(w,\cdot)$ leaves $\sum_x R(w,x)(\mu(x) - \mu'(x))$ unchanged and the shifted reward in $[-\Omega/2, \Omega/2]$, so that sum is at most $(\Omega/2)\|\mu-\mu'\|_1 = \Omega\|\mu-\mu'\|_{\mathrm{TV}}$ in absolute value. Two suprema over w of functions that differ pointwise by at most this amount differ by at most the same amount, which is the first claim. The second follows from $\Delta_Y = \mathbb{E}[\psi(\mu_Y) - \psi(\bar{\mu})] \leq \mathbb{E}|\psi(\mu_Y) - \psi(\bar{\mu})|$.

Binary chain. For the symmetric two-state chain with flip probability p and $r(D) = (1-2p)^D$: conditioning on the delayed state moves the law of X_t by total variation $\frac{1}{2}|r(D)|$ from either value of the conditioner. The Markov property collapses the past to the delayed sample, and it makes the future after t conditionally independent of the past given X_t, so the supremum over $\mathcal{F}_{\geq t}$ reduces to the total variation of the law of X_t alone (the event $\{X_t = 1\}$ attains it); hence $\beta(D) = \frac{1}{2}|r(D)|$. The exact value of the 0/1 prediction objective is $\Delta(D) = \frac{1}{2}|r(D)|$: predict the delayed state if $r > 0$, its complement if $r < 0$; either beats the static $\frac{1}{2}$ by half the correlation magnitude.

Layer-cake decomposition (sketch; the nested joint optimum is not proved). For capacity $a_t \in \{0,\dots,C\}$, threshold indicators $x_t^u = \mathbf{1}[a_t \geq u]$ with $\pi_u = \mathbb{P}(x_t^u{=}1)$, and objectives that are positive combinations of per-layer threshold rewards (goodput minus over-admission surplus in the companion's setting), the objective decomposes additively over layers via $\min(w,a) = \sum_u \mathbf{1}[w \geq u]\, x_t^u$ and $(w-a)^+ = \sum_u \mathbf{1}[w \geq u](1 - x_t^u)$; when the observation is the delayed capacity, the per-layer binary policies (the delayed indicators) are nested, and if that nested policy is jointly optimal, the total advantage over the best comonotone static budget at matched surplus is $\Delta(D) = \sum_u \pi_u r_u(D)$ for layers with $r_u(D) = \mathrm{corr}(x_t^u, x_{t-D}^u) \geq 0$. Joint optimality of the nested policy is the open step.

Battery protocol details. Table 6 lists, for each series, the sample size, cadence, lag grids, selected lag and block length.

Series	N	cadence	lag grid	sel. grid	D^*	block
Two-state Markov	19,874	per step	1–126 (40)	21	76	152
i.i.d. null	19,874	per step	1–126 (40)	21	11	22
LOCATA azimuth	21,452	120 Hz	1–118 (42)	22	118	236
LOCATA VAD	17,708	100 Hz	1–118 (42)	21	1	10
Sperm-whale codas	4,698	per step	1–39 (39)	21	36	72
GOES X-ray flux	9,988	60 s	1–88 (38)	19	55	110
GOES magnetometer	9,988	60 s	1–88 (38)	19	33	66
GOES protons	1,976	300 s	1–39 (39)	20	35	70
Seismic ANMO	12,512	1 s	1–88 (38)	19	5	10
BGP rrc06	9,992	60 s	1–88 (38)	19	67	134

Table 6. Battery protocol details. N: pooled pairs at the least-populated lag. Cadence: sampling interval (whale codas and the synthetic series are indexed per step). Lag grid: registered lags in steps (count); sel. grid: lags entering the in-replicate reselection (every second registered lag, plus D^*). D^*: selected lag; block: moving-block length, $\max(10, 2D^*)$ steps. Binarization thresholds and missing-data handling follow the companion study's preparation scripts, released with the artifact.

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